

Section : Physics

Q.1 A parallel plate capacitor filled with oil of dielectric constant $k = 3$, has a capacitance C . After removing the oil between the plates, the capacitance of the capacitor will become:

1. $3C$
2. $\frac{2}{3}C$
3. C
4. $\frac{1}{3}C$

Options 1. 1
2. 2
3. 3
4. 4

Q.2 If a proton is brought towards an electron from infinity, the potential energy of the system will

1. decrease
2. increase
3. first increase and then decrease
4. remain unchanged

Options 1. 1
2. 2
3. 3
4. 4

Q.3 A conducting wire of length 0.1 m and mass 3×10^{-3} kg, carrying a current of 5 A is suspended freely in a horizontal direction in a uniform magnetic field. The magnitude of magnetic field perpendicular to the length of conductor is:

1. 0.06 T
2. 16.6 T
3. 0.6 T
4. 0.4 T

Options 1. 1
2. 2
3. 3
4. 4

Q.4 The work function of a photosensitive material is 4.0 eV. The longest wavelength of light that can cause electron emission from the material is (approximately)

1. 3100 nm
2. 966 nm
3. 31 nm
4. 310 nm

Options 1. 1
2. 2
3. 3
4. 4

Q.5 A battery of emf 3 V and internal resistance r is connected in series with a resistor of 55Ω . An ammeter of resistance 2Ω connected in circuit reads 50 mA. The value of r is

1. 2Ω
2. 3Ω
3. 4Ω
4. 1Ω

Options 1. 1
2. 2
3. 3
4. 4

Q.6 Choose the correct relation between focal length f and radius of curvature R of a convex mirror from the following:

1. $f = +R$
2. $f = -R$
3. $f = -0.5 R$
4. $f = +0.5 R$

Options 1. 1
2. 2
3. 3
4. 4

Q.7 For a series LCR circuit, $R = X_L = 3X_C$.

The impedance of the circuit and the phase difference between voltage V and current I , respectively, will be:

1. $3R\sqrt{13} \Omega$ and $\tan^{-1}\left(\frac{3}{2}\right)$
2. $3R\sqrt{13} \Omega$ and $\tan^{-1}\left(\frac{2}{3}\right)$
3. $\frac{R\sqrt{13}}{3} \Omega$ and $\tan^{-1}\left(\frac{2}{3}\right)$

4. $\frac{R\sqrt{13}}{3}\Omega$ and $\tan^{-1}\left(\frac{3}{2}\right)$

- Options 1. 1
2. 2
3. 3
4. 4

Q.8 Two lenses of focal lengths, 20 cm and - 25 cm, respectively are placed in contact with each other. The power of the combination of lenses is

1. 1D
2. 10 D
3. -5 D
4. -1 D

- Options 1. 1
2. 2
3. 3
4. 4

Q.9 An alpha particle is moving in a magnetic field $(3\hat{i} + 2\hat{j})$ T with the velocity $(7 \times 10^5\hat{i})$ m/s. The magnetic force acting on the particle is:

1. $3.4 \times 10^{-13} \hat{k}$ N
2. $4.48 \times 10^{-13} \hat{k}$ N
3. $3.4 \times 10^{-11} \hat{k}$ N
4. $4.48 \times 10^{-11} \hat{k}$ N

- Options 1. 1
2. 2
3. 3
4. 4

Q.10 The 'stopping potential' in photoelectric emissions

1. depends on both intensity and frequency of incident radiation.
2. depends on frequency and not on intensity of incident radiation.
3. depends on intensity and not on frequency of incident radiation.
4. does not depend on intensity and frequency of incident radiation.

- Options 1. 1
2. 2
3. 3
4. 4

Q.11 An electron in the ground state goes to the first excited state in the hydrogen atom. Arrange the following energies in decreasing order

- (A) Total energy in ground state
(B) Kinetic energy in ground state
(C) Potential energy in ground state
(D) Total energy in first excited state

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (A), (C), (D)
3. (B), (D), (A), (C)
4. (B), (C), (D), (A)

- Options 1. 1
2. 2
3. 3
4. 4

Q.12 The Young's double slit experiment is performed with monochromatic blue, green, yellow and red lights, respectively.

Arrange the distance (x) of 4th maxima from the central maximum for all the colors in decreasing order:

(A) X_{red}

(B) X_{blue}

(C) X_{green}

(D) X_{yellow}

Choose the correct answer from the options given below:

1. (A), (D), (C), (B)
2. (B), (D), (C), (A)
3. (B), (C), (D), (A)
4. (B), (D), (A), (C)

- Options 1. 1
2. 2
3. 3
4. 4

Q.13 Match List-I with List-II

List-I

List-II

Physical Quantity (According to Rutherford Model) Formula

(A) Total energy of the electron in hydrogen atom

$$(I) \frac{e^2}{8\pi\epsilon R}$$

(B) Kinetic energy of the electron in hydrogen atom

$$(II) \frac{-e^2}{8\pi\epsilon R}$$

(C) Radius of the circular orbit around the nucleus

$$(III) \frac{-e^2}{4\pi\epsilon R}$$

(D) Potential energy of the electron in hydrogen atom

$$(IV) \frac{e^2}{4\pi\epsilon m v^2}$$

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
2. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Identify the correct statements among the following:

(A) The value of the electrostatic force constant, k depends on the medium between the two charges and on the choice of system of units.

(B) For two positively charged bodies, one more highly charged than the other, the force between them may be attractive.

(C) The charge on a body does change with its velocity.

(D) An electric dipole is a pair of two equal and similar charges placed at a very small distance.

Choose the correct answer from the following options:

1. (A) and (C) only
2. (A) and (B) only
3. (C) and (D) only
4. (A) only

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Match List-I with List-II

List-I

List-II

Electromagnetic Wave Application

(A) Microwaves

(I) Photographs during foggy season and night vision cameras

(B) X-rays

(II) Radar

(C) Gamma rays

(III) Photographs internal parts of human body

(D) Infrared Rays

(IV) Radiography for malignant tumors in cancer treatment

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
4. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Two equal and similar charges 3 cm apart in air repel each other with a force of 1.6 N.

What will be the magnitude of each charge?

1. $12 \times 10^{-7} \text{ C}$
2. $3 \times 10^{-7} \text{ C}$
3. $4 \times 10^{-7} \text{ C}$
4. $1.33 \times 10^{-8} \text{ C}$

Options 1. 1

2. 2
3. 3
4. 4

Q.17 In a Young's double slit experiment, the two parallel slits are 1 mm apart and a screen is placed 1 m away from the

slits. When the light of wavelength 600 nm is used, the fringe width is

1. 0.4 mm
2. 0.5 mm
3. 0.6 mm
4. 0.7 mm

Options 1. 1

2. 2
3. 3
4. 4

Q.18 A piece of silver and another of silicon are cooled from 15°C to 3°C . What will be the effect on resistance of each?

1. Silver: increases ; Silicon : increases
2. Silver: decreases ; Silicon : increases
3. Silver: increases ; Silicon : decreases
4. Silver: decreases ; Silicon : decreases

Options

1. 1
2. 2
3. 3
4. 4

Q.19 In a diffraction from a single slit of width 0.30 mm , first order maxima is found with light of wavelength $6000\text{ \AA}$. The angular spread of first order maxima on either side of the central maxima is:

1. $6 \times 10^{-3}\text{ rad}$
2. $3 \times 10^{-3}\text{ rad}$
3. $2 \times 10^{-3}\text{ rad}$
4. $4 \times 10^{-3}\text{ rad}$

Options

1. 1
2. 2
3. 3
4. 4

Q.20 Seven lead accumulators, each of emf 2 V , and internal resistance $0.07\text{ }\Omega$, are connected in series with an external resistance $9.51\text{ }\Omega$. The current in the circuit is

1. 1.4 A
2. 0.5 A
3. 0.7 A
4. 10 A

Options

1. 1
2. 2
3. 3
4. 4

Q.21 Two point charges of 2 C each are separated from each other by a distance of 1 m in water ($K = 80$).

The force between the charges is:

1. $4.5 \times 10^9\text{ N}$
2. $4.5 \times 10^8\text{ N}$
3. $2.5 \times 10^9\text{ N}$
4. $2.5 \times 10^8\text{ N}$

Options

1. 1
2. 2
3. 3
4. 4

Q.22 Which of the following statement(s) is/are incorrect?

- (A) The Kirchhoff's laws are equally applicable in alternating current circuit and direct current circuits,
- (B) It is not necessary that the wire stretched on the meter bridge must have a uniform cross-sectional area.
- (C) The internal resistance of the cell does not affect the reading of the null point in metre bridge.
- (D) Electrical potential has no direction, the electrical potential gradient has a direction.

1. (A) and (B) only
2. (B) only
3. (A) and (C) only
4. (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.23 A charged particle moving with a certain velocity enters a uniform electric field perpendicular to its direction.

In this

context, which of the following statements are correct?

- (A) Force acting on the particle is constant
 - (B) Acceleration of the particle is constant
 - (C) Velocity of the particle is constant
 - (D) Displacement of the particle follows a parabolic path
- Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.24 Match List-I with List-II

List-I

List-II

- (A) Current (I) Opposition to flow of current
- (B) Resistance (II) Rate of flow of charge
- (C) Resistivity (III) Work done per unit charge

(D) EMF (IV) Resistance per unit length $\times$ unit cross sectional area

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.25 Identify the incorrect statement(s) among the following:

When a dielectric medium is introduced between the plates of a charged capacitor, the electric field inside the dielectric changes due to polarisation. E_0 is the electric field without dielectric and E_p is the electric field due to polarisation.

(A) If a plate of dielectric constant k , thickness t is placed between the plates of a charged capacitor of area A

of the plate and having distance d , between the plates then the new capacitance will be $\frac{A\epsilon_0}{d - t(1 + 1/k)}$

(B) The electric field between plates of the charged capacitor when the dielectric is introduced will be $E = (E_0 - E_p)$

(C) Dielectric constant of the dielectric placed between plates of charged capacitor will be $k = \frac{E_0}{E_0 - E_p}$

(D) Dielectric strength of the dielectric is the maximum value of the electric field which may be applied before its breakdown.

Choose the correct answer from the options given below:

1. (A) only
2. (B) and (D) only
3. (C) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.26 A current of 5 A is passing through a metallic wire of cross-sectional area $5 \times 10^{-6} \text{ m}^2$.

If the density of the charge carriers in the wire is $5 \times 10^{26} \text{ m}^{-3}$, the drift velocity of the electrons is:

1. $1.25 \times 10^{-2} \text{ m/s}$
2. $1.75 \times 10^{-2} \text{ m/s}$
3. $2.25 \times 10^2 \text{ m/s}$
4. $2.50 \times 10^2 \text{ m/s}$

Options 1. 1

2. 2
3. 3
4. 4

Q.27 The parameter to be increased in order to increase the angular magnification of a simple microscope is

1. aperture of the lens
2. size of the object
3. power of the lens
4. focal length of the lens

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Two point charges $+200 \mu\text{C}$ are placed at a distance of 2 mm from each other in a uniform electric field of 10^5 N/C , such that the line joining the negative to positive charge makes an angle 30° with the field.

The torque acting on the system is:

1. 0.02 Nm
2. 0.2 Nm
3. 2.02 Nm
4. 0.002 Nm

Options 1. 1

2. 2
3. 3
4. 4

Q.29 The energy equivalent of 10 g of substance is:

1. $3 \times 10^8 \text{ J}$
2. $9 \times 10^8 \text{ J}$
3. $9 \times 10^{14} \text{ J}$
4. $9 \times 10^{13} \text{ J}$

Options 1. 1

2. 2
3. 3

Q.30 Read the following statements carefully:

- (A) The average power in an inductor over one complete cycle is zero.
 (B) Self-inductance plays the role of inertia.
 (C) Mutual inductance of a pair of coils depends on their separation as well as their relative orientation.
 (D) The capacitive reactance increases with increase in frequency of applied a. c.

Choose the **correct** answer from the options given below:

1. (A), (B) and (C) only
2. (A), (C) and (D) only
3. (B), (C) and (D) only
4. (A), (B), (C) and (D)

Options 1. 1

2. 2

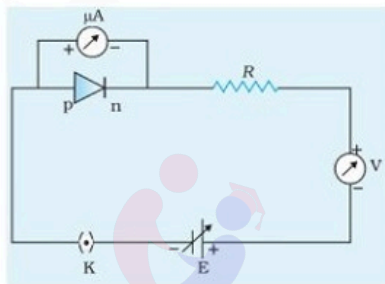
3. 3

4. 4

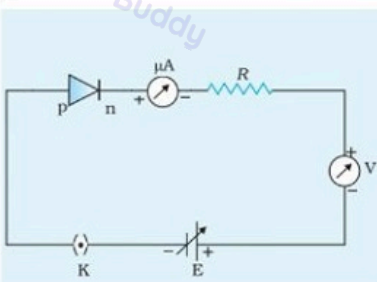
Q.31

Choose the correct circuit for the forward bias of p-n junction diode:

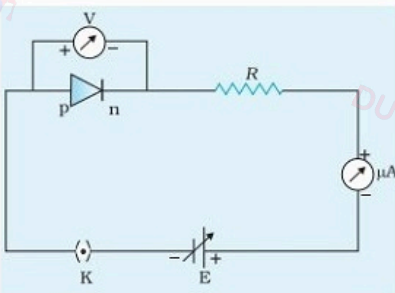
(A)



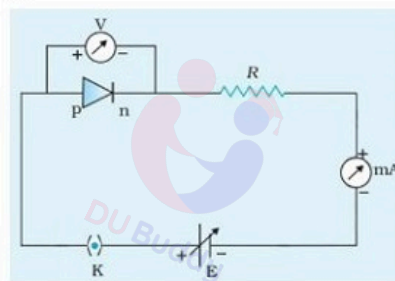
(B)



(C)



(D)



Choose the **correct** answer from the options given below:

1. (A) and (D) only
2. (C) and (D) only
3. (A) and (B) only
4. (D) only

Options 1. 1

2. 2

3. 3

Q.32 A bar magnet is cut into two equal halves along a plane parallel to the magnetic axis. The one parameter which remains unchanged is

1. intensity of magnetisation
2. magnetic moment
3. pole strength
4. Magnetic field lines

Options 1. 1
2. 2
3. 3
4. 4

Q.33 Choose the correct phase relationship between the current and the applied voltage in an a.c. circuit.
(A) In a pure resistive circuit, the current is in phase with the applied voltage.
(B) In a pure inductive circuit, the current lags behind the voltage by a phase angle of 90° .
(C) In a pure inductive circuit, the current leads the voltage by a phase angle of 90° .
(D) In a pure resistive circuit, the current is not in phase with the applied voltage.
Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A) and (C) only
3. (C) and (D) only
4. (B) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.34 The Earth's magnetic field at a given point is $0.5 \times 10^{-5} \text{ Wb/m}^2$.
This field is to be annulled by magnetic induction at the center of a circular conducting loop of radius 5 cm.
The current required to be flown in the loop is nearly:

1. 4 A
2. 0.2 A
3. 0.4 A
4. 20 A

Options 1. 1
2. 2
3. 3
4. 4

Q.35 The electrical resistance to the diffusion current in the depletion layer in a p-n junction is large due to the presence of

1. strong electric field
2. large number of charge carriers
3. electrons as charge carriers
4. holes as charge carriers

Options 1. 1
2. 2
3. 3
4. 4

Q.36 What will be the emf induced in a square shaped coil of side 0.2 m having 200 turns, placed perpendicular to a magnetic field which is increasing at the rate of 0.5 T/s ?

1. 20 V
2. 2.0 V
3. 4.0 V
4. 1.6 V

Options 1. 1
2. 2
3. 3
4. 4

Q.37 Identify the correct statements among the following:
(A) An electromagnetic wave travels in vacuum like sound waves.
(B) An ultraviolet wave is not an electromagnetic wave.
(C) An electromagnetic wave can travel from earth to the planet, saturn.
(D) An electromagnetic wave may have less velocity in a medium than in vacuum.
Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A) and (C) only
3. (C) and (D) only
4. (B) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

- Q.38 The self-inductance (L) of a long solenoid on doubling its number of turns and length, keeping other factors constant, will become
1. $L/2$
 2. L
 3. $2L$
 4. $4L$

Options 1. 1
2. 2
3. 3
4. 4

- Q.39 For a photoelectric effect
- (A) the photoelectric current is directly proportional to the intensity of the incident radiation.
- (B) the maximum kinetic energy of the emitted photoelectrons increases linearly with the frequency of the incident radiation.
- (C) for a given frequency of the incident radiation, the stopping potential increases linearly with the intensity of the incident radiation.
- (D) the photoelectric emission is an instantaneous process without any apparent time lag.
- Choose the correct answer from the options given below:
1. (A), (B) and (D) only
 2. (A), (B) and (C) only
 3. (A), (B), (C) and (D)
 4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

- Q.40 Identify the correct statement(s) among the following:
- (A) The time taken by a light wave is 5×10^{-9} s to travel a thickness of 1 cm of glass having refractive index 1.5.
- (B) Two similar waves of amplitude 3 mm and 5 mm reach a point in the same phase. The resultant amplitude will be 8 mm.
- (C) Two waves of amplitude 2 mm each reach a point with a phase difference of $\pi/2$. The resultant amplitude is zero.
- (D) If one of the slits is closed in Young's double slit experiment, the interference pattern will disappear.
- Choose the correct answer from the options given below:
1. (A) and (D) only
 2. (A), (B) and (D) only
 3. (B) and (D) only
 4. (D) only

Options 1. 1
2. 2
3. 3
4. 4

- Q.41 Which of the following statement(s) is/are incorrect?
- (A) A current carrying loop may act as a magnetic dipole whose magnetic moment may depend on its area and the number of turns.
- (B) A linear solenoid carrying current, may act as a bar magnet.
- (C) The magnetic lines of forces due to the current carrying solenoid makes closed curves like in a bar magnet.
- (D) The magnetic field inside a solenoid is perpendicular to its axis.
- Choose the correct answer from the given options:
1. (A) and (C) only
 2. (B) and (D) only
 3. (C) and (D) only
 4. (D) only

Options 1. 1
2. 2
3. 3
4. 4

- Q.42 A charged particle moves along a circular path in a uniform magnetic field of 0.04 T with a velocity of 2×10^5 m/s. The charge to mass ratio of the particle is 5×10^7 C/kg. The radius of the circular path of the charged particle is:
1. 1000 cm
 2. 10 cm
 3. 50 cm
 4. 100 cm

Options 1. 1
2. 2
3. 3
4. 4

- Q.43 Light with an energy flux of 18 W/cm^2 falls at normal incidence on a non-reflecting surface of area 20 cm^2 . If the light falls for 30 minutes on the surface, find the total momentum delivered to the surface.
1. $10^{-3} \text{ Kg m s}^{-1}$
 2. 2.16 Kg m s^{-1}
 3. $4.32 \times 10^{-3} \text{ Kg m s}^{-1}$

4. $2.16 \times 10^{-3} \text{ Kg m s}^{-1}$

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Identify the correct statement(s) among the following:

(A) If three electrical cells each of emf, E and internal resistance r are connected in series to a resistance R , the current drawn from them is $3E/(R+3r)$.

(B) If these three cells are connected in parallel combination, across resistance R then the current drawn will be $E/(3R+r)$.

(C) If the two cells are connected in series and then one in parallel which are then connected across resistance R , then the current drawn will be $2E/(2R+2r)$.

(D) In a mixed grouping of cells, current will be maximum when total external resistance is equal to total equivalent internal resistance.

Choose the correct answer from the options given below

1. (A) only

2. (A) and (D) only

3. (A), (C) and (D) only

4. (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.45 The angular momentum of an electron moving in the circular path of diameter D in a uniform magnetic field B_0 is:

1. zero

2. $\frac{eB_0D}{2}$

3. $\frac{eB_0D^2}{2}$

4. $\frac{eB_0D^2}{4}$

Options 1. 1

2. 2

3. 3

4. 4

Q.46 A nucleus disintegrates into two nuclear fragments which move apart, having their velocities in the ratio 1:3. The ratio of their nuclear sizes, respectively, will be

1. $\sqrt[3]{3} : 1$

2. $\sqrt{3} : 1$

3. $1 : \sqrt[3]{3}$

4. $1 : \sqrt{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.47 In a circuit, resistance R_1 is connected in series with the parallel combination of resistances R_2 and R_3 . The currents passing through the resistances R_1 , R_2 and R_3 are I_1 , I_2 and I_3 , respectively.

What will be the ratio $\frac{I_3}{I_1}$ of currents in terms of the resistances connected in the circuit?

1. $\frac{R_2 + R_3}{R_1}$

2. $\frac{R_2}{R_2 + R_3}$

3. $\frac{R_3}{R_1 + R_2}$

4. $\frac{R_1}{R_2 + R_3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.48 A series LCR circuit is connected to a 230 V, 50 rad/s source. Given $L = 5 \text{ H}$, $C = 80 \mu\text{F}$ and $R = 40 \Omega$. Impedance of the circuit is

1. $40\ \Omega$
2. $50\ \Omega$
3. $100\ \Omega$
4. $250\ \Omega$

Options

1. 1
2. 2
3. 3
4. 4

Q.49 The current in a coil falls from 10.0 A to 0.0 A in 0.1 s . If an average emf of 100 V is induced in the coil, then the self-inductance of the coil is.

1. 0.1 H
2. 1 H
3. 10 H
4. 5 H

Options

1. 1
2. 2
3. 3
4. 4

Q.50 The far point of a myopic person is 80 cm . The power of corrective lens will be

1. -1.25 D
2. 1.25 D
3. -2.5 D
4. 1.5 D

Options

1. 1
2. 2
3. 3
4. 4